VIT – Vellore Institute of Technology

Vellore – 632014, Tamil Nadu, India

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Fall Semester 2026-2027

Database Systems (BCSE307L) – Project Report

[PROJECT TITLE]

Programme	B.Tech. (All Branches)
Slot	L11+L12 and L43+L44
Course Name & Code	Database Systems & BCSE307L
Class Number	[VL2026270104071 / VL2026270104074 – keep yours]
Faculty Name	Dr. Deepika J

Team Members

S.No	Name	Reg. No.	Role
1	*[Name]*	*[Reg. No.]*	*[Role]*
2	*[Name]*	*[Reg. No.]*	*[Role]*
3	*[Name]*	*[Reg. No.]*	*[Role]*
4	*[Name]*	*[Reg. No.]*	*[Role]*

Project URLs and Resources

Resource	URL
GitHub Repository	*[URL]*
Live Deployment	*[URL]*
Docker Hub Image	*[URL]*
Frontend Framework / Docs	*[URL]*
Backend Framework / Docs	*[URL]*
ORM / Docs	*[URL]*
Database / Docs	*[URL]*
Authentication Provider (OAuth/JWT)	*[URL]*
Datasets / APIs (if any)	*[URL]*
Other tools	*[URL]*

1. Problem Statement and Project Planning

1.1 Project Title

[Write your project title.]

1.2 Problem Statement

[Write in your own words.]

1.3 Objectives

- *[Objective 1]*
- *[Objective 2]*
- *[Objective 3]*

1.4 Scope

[What is included and what is not.]

1.5 Team Members and Responsibilities

Name	Reg. No.	Responsibilities
[Name]	*[Reg. No.]*	*[Responsibilities]*
[Name]	*[Reg. No.]*	*[Responsibilities]*

Name	Reg. No.	Responsibilities
[Name]	*[Reg. No.]*	*[Responsibilities]*

1.6 Technology Stack

Layer	Technology
Frontend	*[ ]*
Backend	*[ ]*
Database	*[ ]*
ORM	*[ ]*
Authentication	*[ ]*
Deployment	*[ ]*

1.7 Major Deliverables

- *[Deliverable 1]*
- *[Deliverable 2]*

1.8 Project Timeline (Gantt Chart)

[Insert screenshot here]

Figure 1: Project Gantt chart

2. Database Design

2.1 Entity Relationship (ER) Diagram

[Explain entities and relationships.]

[Insert screenshot here]

Figure 2: ER Diagram

2.2 Relational Schema

[Insert screenshot here]

Figure 3: Relational schema derived from the ER diagram

2.3 Data Dictionary

Table	Column	Data Type	Key	Constraints
[ ]	*[ ]*	*[ ]*	*[PK/FK]*	*[NOT NULL, UNIQUE, CHECK, DEFAULT]*
[ ]	*[ ]*	*[ ]*	*[ ]*	*[ ]*

2.4 Keys and Constraints

[Explain PKs, FKs and constraints used.]

[Insert screenshot here]

Figure 4: Constraints in the CREATE TABLE statements

2.5 Normalization (up to 3NF)

[Show UNF → 1NF → 2NF → 3NF for an example.]

[Insert screenshot here]

Figure 5: Normalization steps

2.6 Indexes

[Which indexes and why.]

[Insert screenshot here]

Figure 6: CREATE INDEX statements and index list

3. Database Implementation

3.1 Database Creation and Tables

[Explain in your own words. Show input, processing and output.]

[Insert screenshot here]

Figure 7: List of all tables in the database

3.2 CRUD Operations

[Explain in your own words. Show input, processing and output.]

[Insert screenshot here]

Figure 8: CRUD through the UI and resulting database rows

3.3 Joins

[Explain in your own words. Show input, processing and output.]

[Insert screenshot here]

Figure 9: Join query and its output

3.4 Aggregate Queries

[Explain in your own words. Show input, processing and output.]

[Insert screenshot here]

Figure 10: Aggregate query and its output

3.5 Views

[Explain in your own words. Show input, processing and output.]

[Insert screenshot here]

Figure 11: View creation and SELECT from the view

3.6 Transactions

[Explain in your own words. Show input, processing and output.]

[Insert screenshot here]

Figure 12: Transaction commit and rollback

3.7 Stored Procedure

[Explain in your own words. Show input, processing and output.]

[Insert screenshot here]

Figure 13: Stored procedure definition and execution

3.8 Trigger

[Explain in your own words. Show input, processing and output.]

[Insert screenshot here]

Figure 14: Trigger definition and firing demonstration

3.9 ORM Integration

[Explain in your own words. Show input, processing and output.]

[Insert screenshot here]

Figure 15: ORM models/entities and repository or service code

4. Application Development

4.1 Login Page

[Explain.]

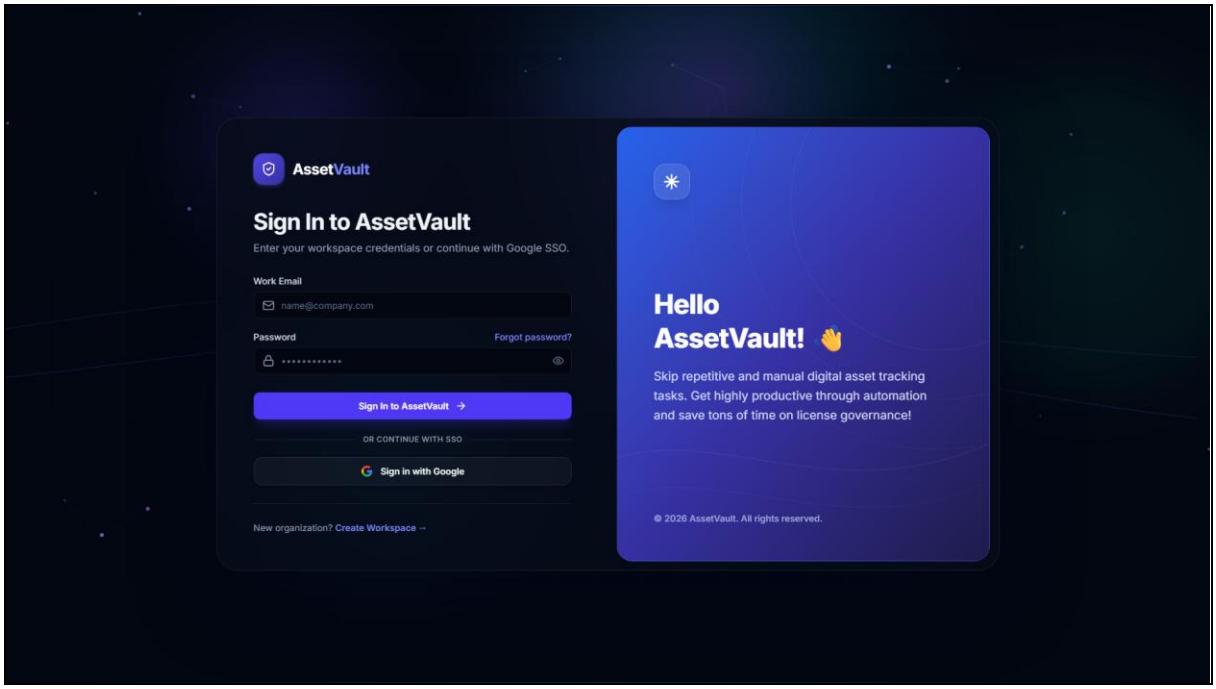


Figure 16: Login page

4.2 Dashboard

[Explain.]

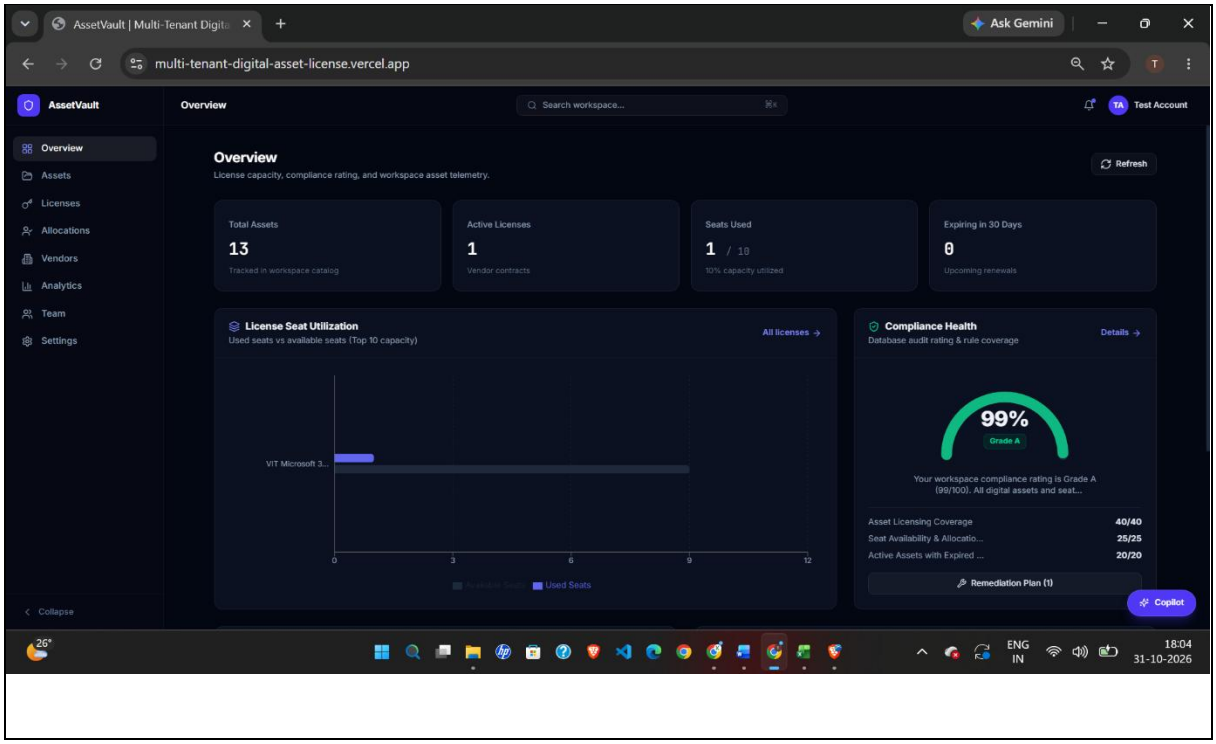


Figure 17: Dashboard

4.3 Search, Filtering and Pagination

[Explain.]

[Insert screenshot here]

Figure 18: Search, filter and pagination in use

4.4 Form Validation

[Explain.]

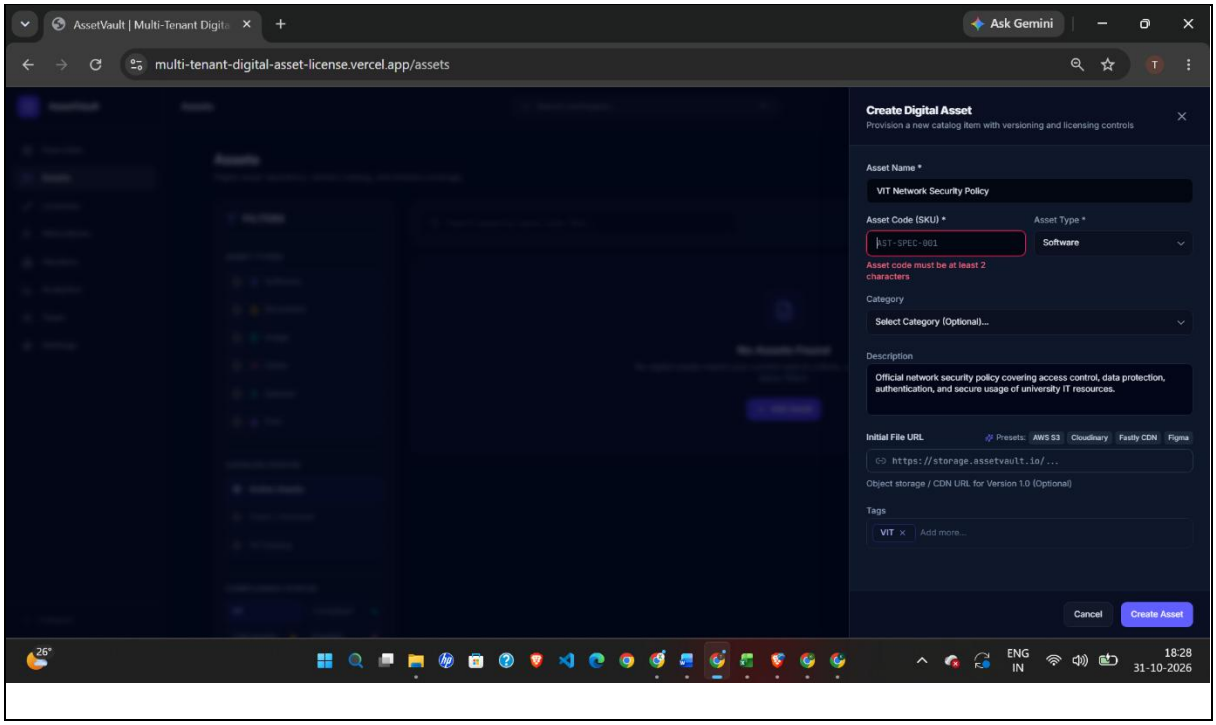


Figure 19: Form validation errors

4.5 Exception Handling

[Explain.]

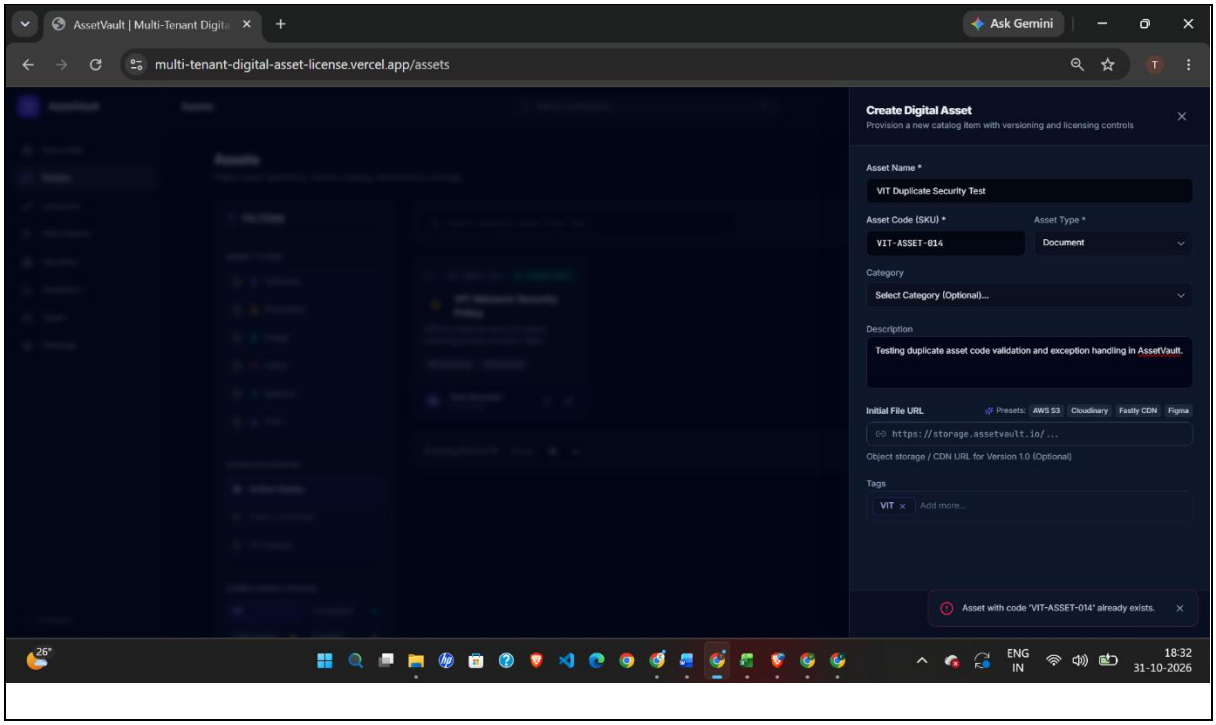
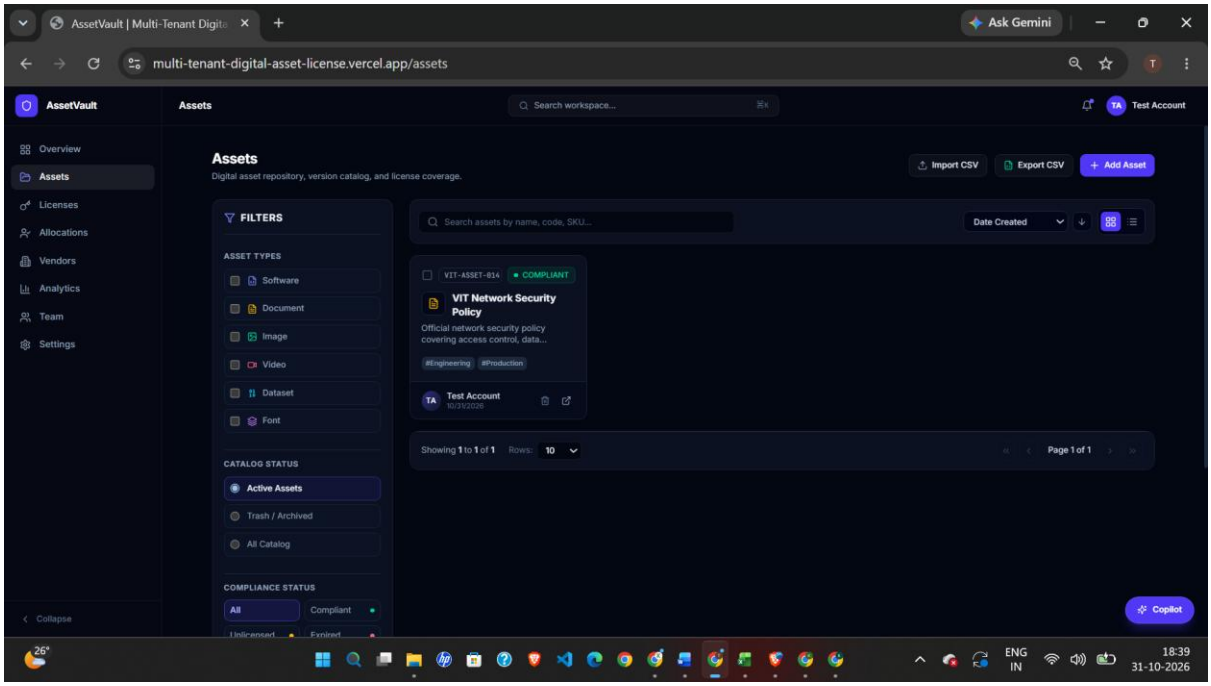


Figure 20: Error handling (e.g. 404 / duplicate entry)

4.6 Responsive UI

[Explain.]



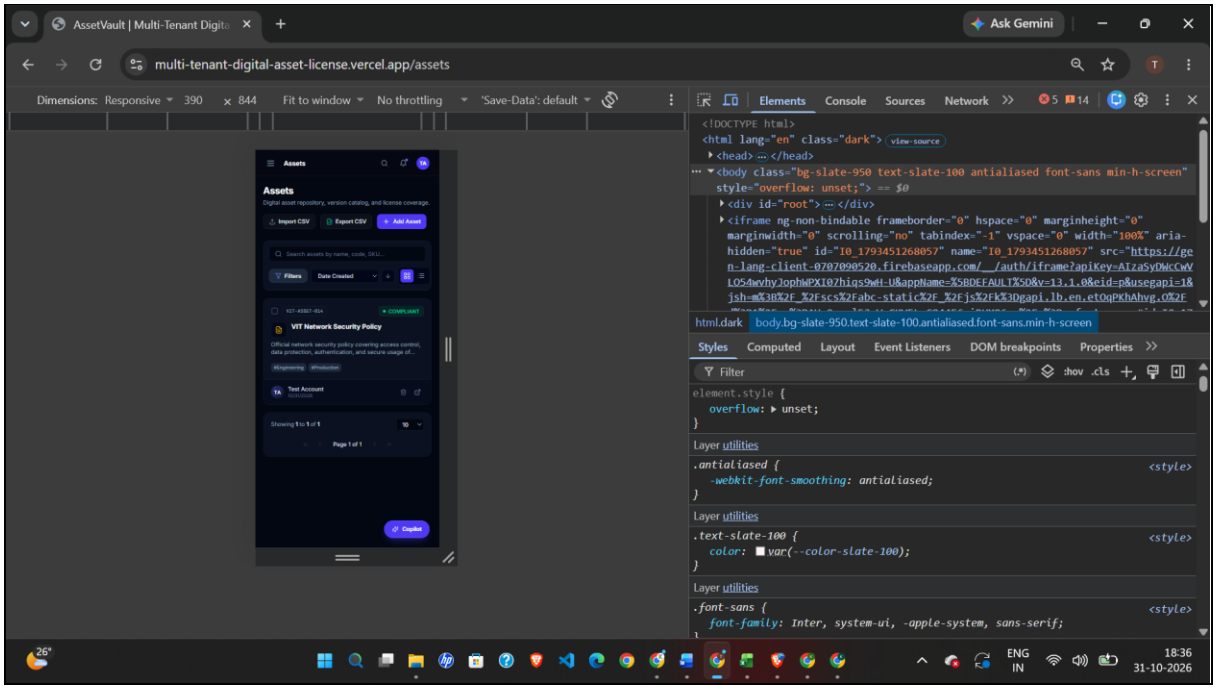
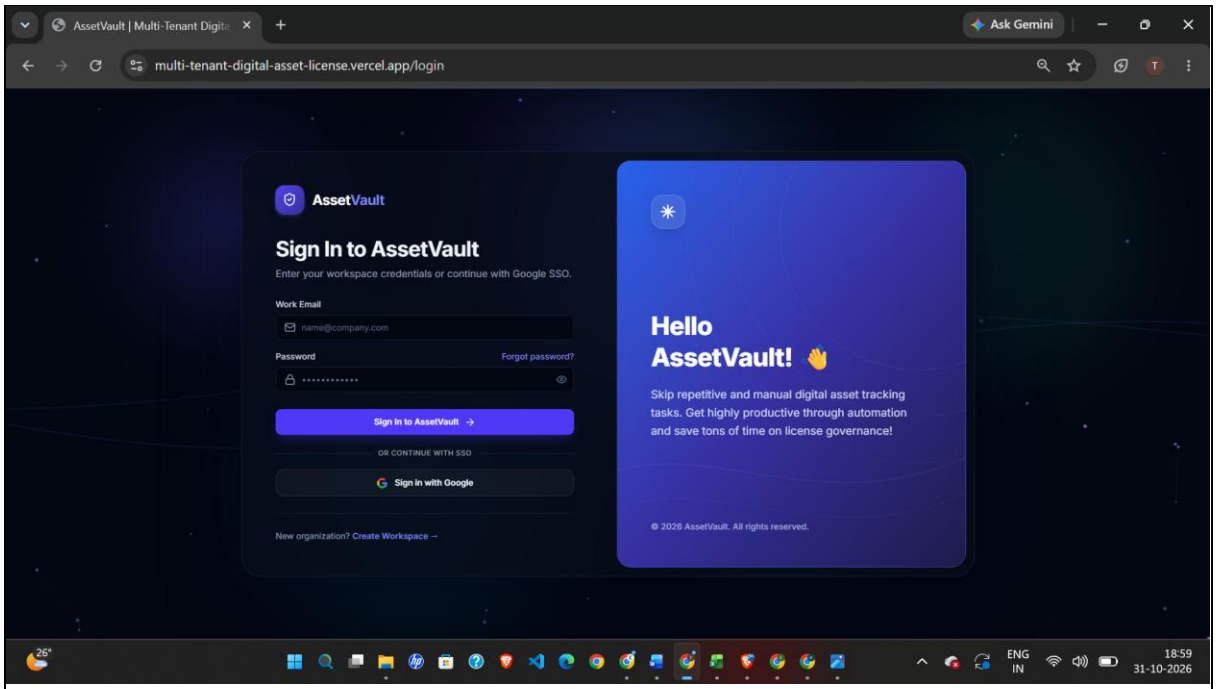


Figure 21: Desktop and mobile views

5. Authentication, Authorization and Database Security

5.1 Authentication (Login, Logout, Token/Session Handling)

[Explain.]



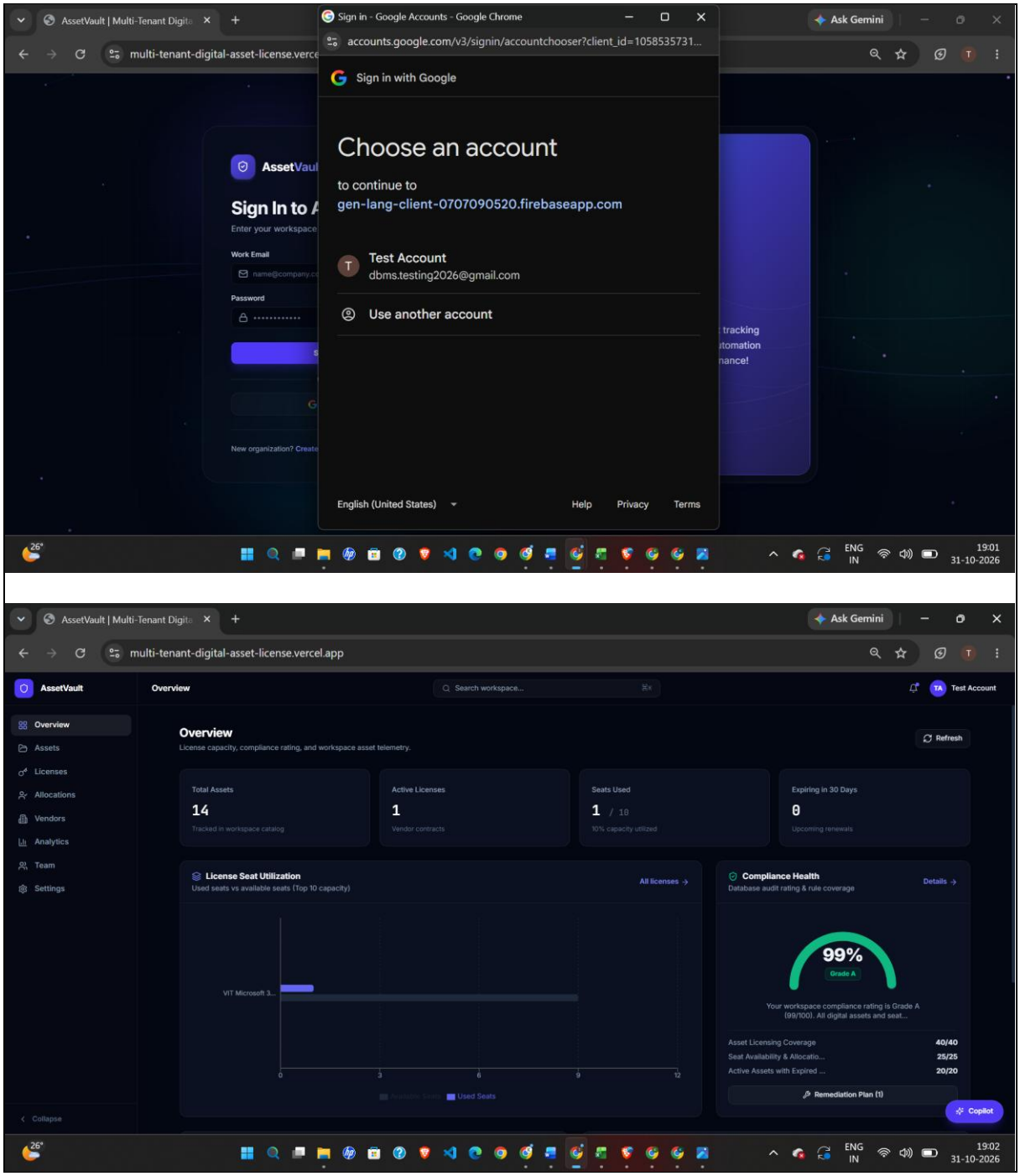


Figure 22: Login, logout and token/session

5.2 Retrieval of Authenticated User Information

[Explain.]

[Insert screenshot here]

Figure 23: Authenticated user details

5.3 Role-Based Access Control (Single Users Table)

[Explain.]

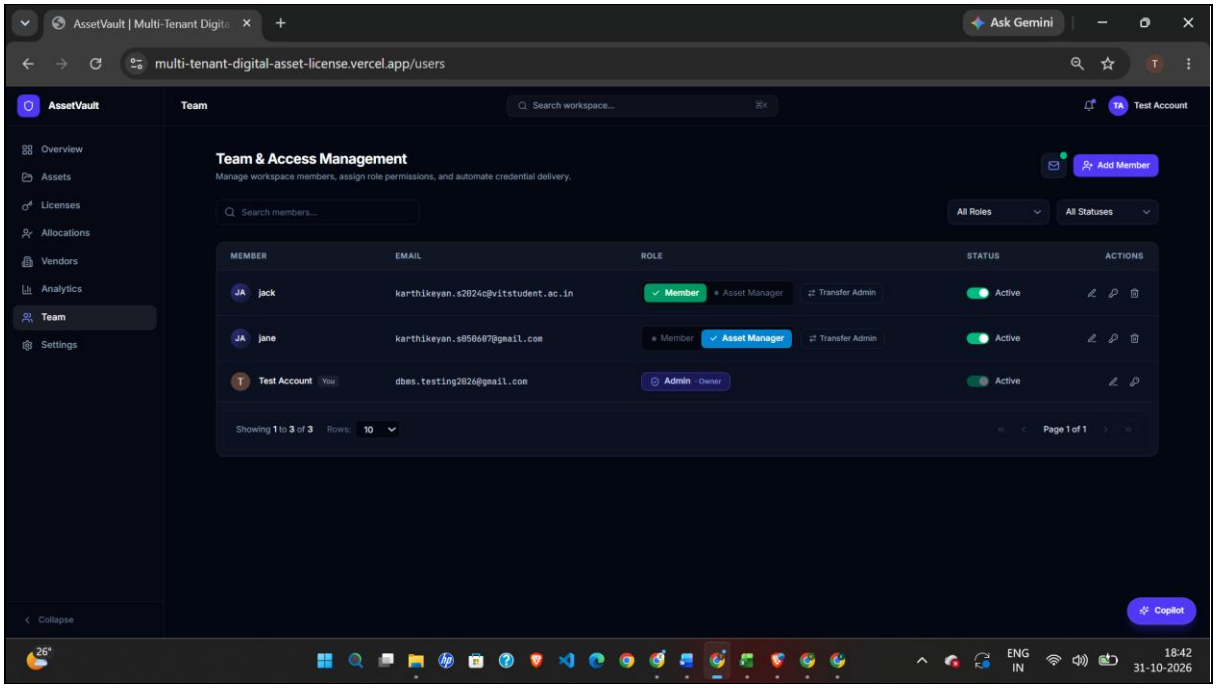


Figure 24: Users table with role column; role-based menu, page and API access

5.4 Unauthorized Access Blocked

[Explain.]

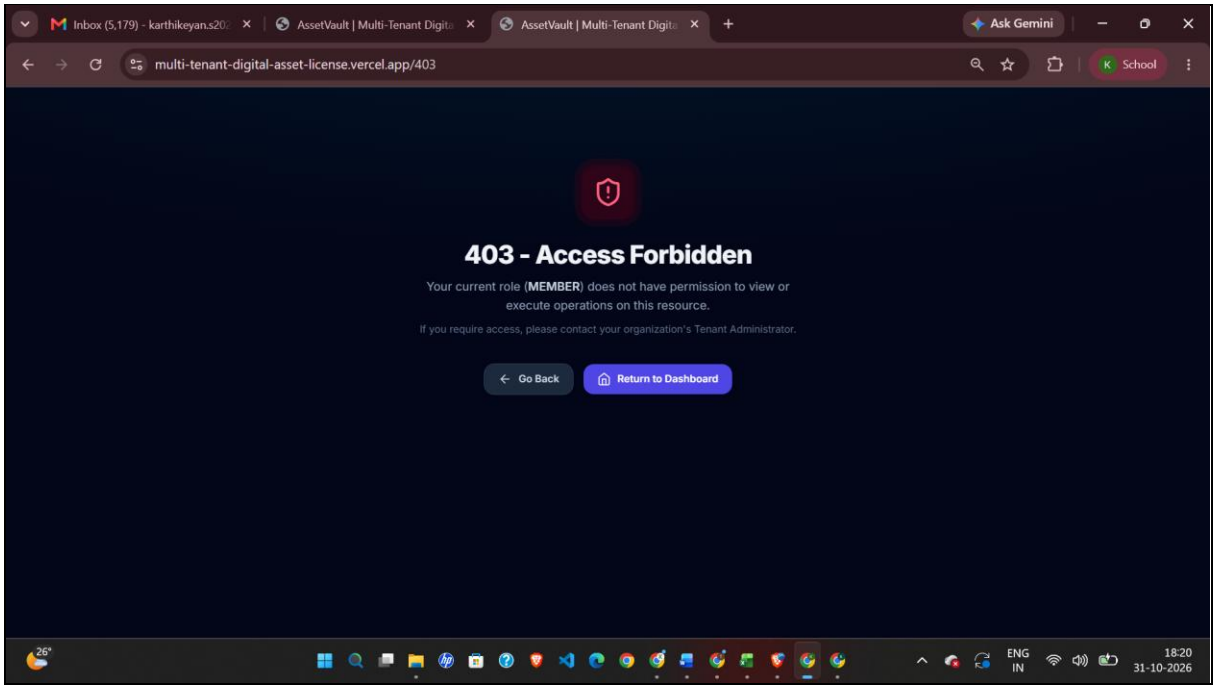


Figure 25: API response (403) for wrong role

5.5 Password Hashing

[Explain.]

[Insert screenshot here]

Figure 26: Hashed passwords in the database

5.6 SQL Injection Prevention

[Explain.]

[Insert screenshot here]

Figure 27: ORM parameterized queries

5.7 Secrets Management

[Explain.]

[Insert screenshot here]

Figure 28: .env usage and .gitignore

6. Professional Development Practices

6.1 GitHub Repository, Branches and Pull Requests

[Explain branching strategy.]

[Insert screenshot here]

Figure 29: GitHub repository showing main and dev branches

6.2 Individual Commit History

Member / ID	Commit ID	Branch	Date	Commit Message
[ ]	*[ ]*	*[ ]*	*[ ]*	*[ ]*
[ ]	*[ ]*	*[ ]*	*[ ]*	*[ ]*
[ ]	*[ ]*	*[ ]*	*[ ]*	*[ ]*

[Insert screenshot here]

Figure 30: git log output

6.3 .gitignore and README

[Insert screenshot here]

Figure 31: .gitignore and README.md

6.4 Database Backup and Restore

[Native tool commands used.]

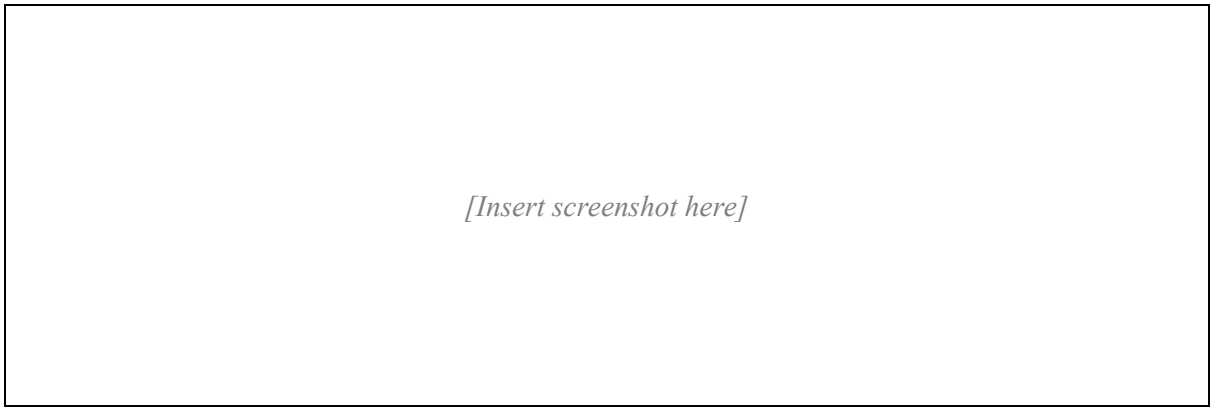


Figure 32: Backup command and output

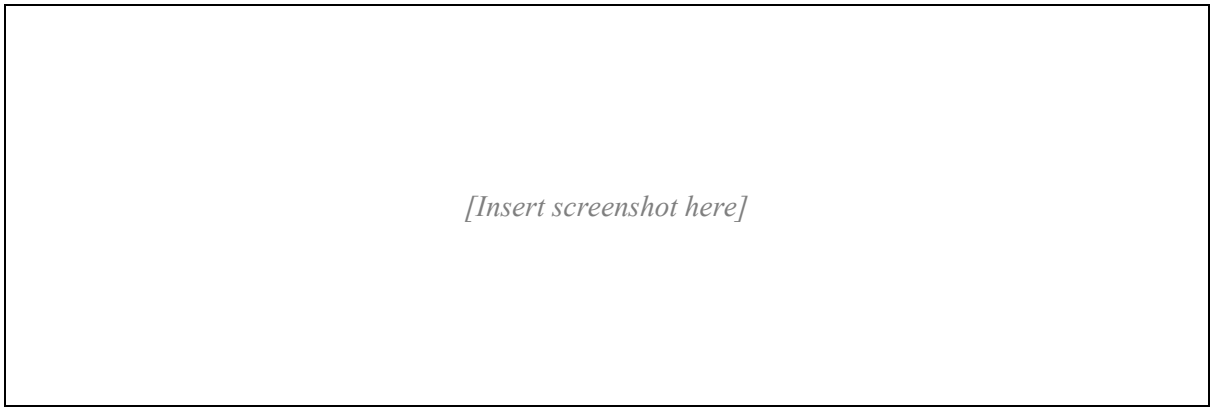


Figure 33: Restore command and output

6.5 Docker and Docker Hub

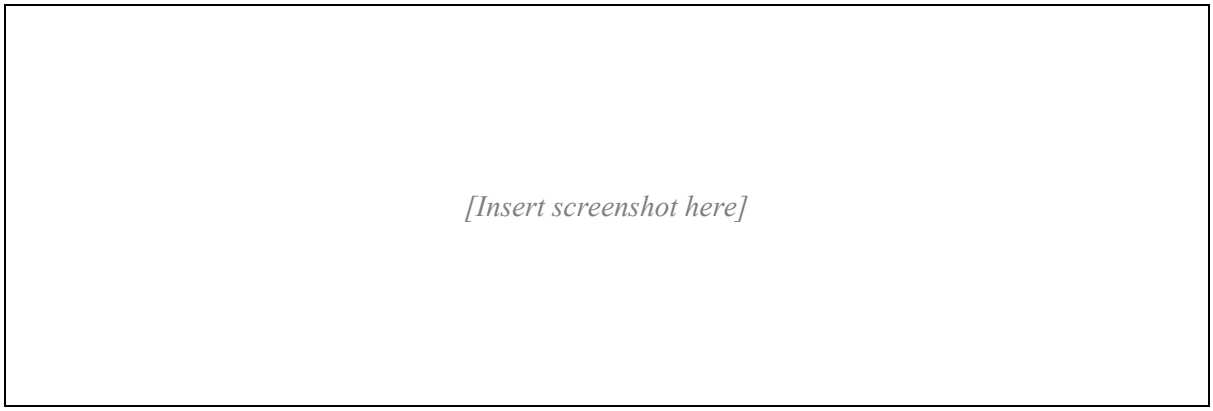


Figure 34: Dockerfile and running container

[Insert screenshot here]

Figure 35: Docker Hub repository

6.6 Cloud Deployment

[Live URL:]

[Insert screenshot here]

Figure 36: Deployed application in browser

6.7 Bonus Features (if any)

[CI/CD, custom domain, audit logging, testing, monitoring, etc.]

[Insert screenshot here]

Figure 37: Bonus feature evidence

7. Project Execution (Inputs, Processing, Outputs)

[Walk through one end-to-end flow: input → processing → output.]

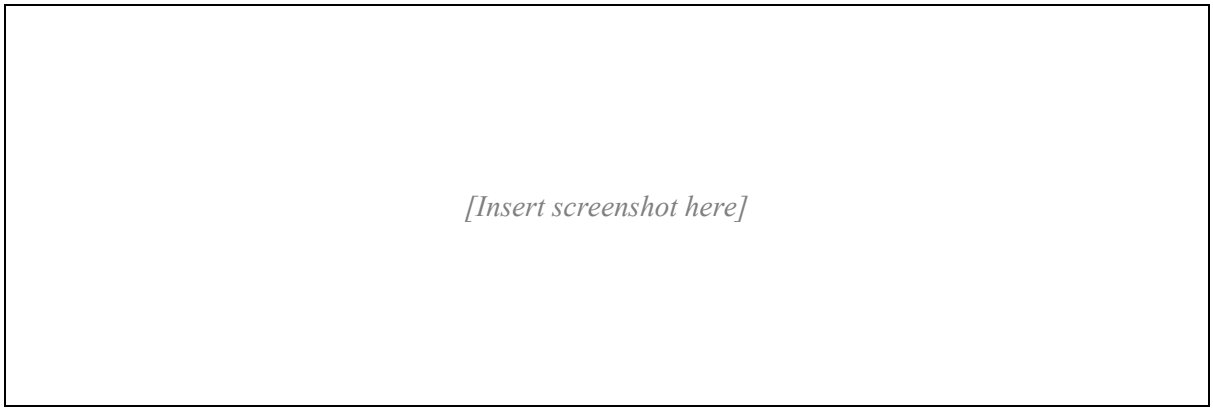


Figure 38: Input

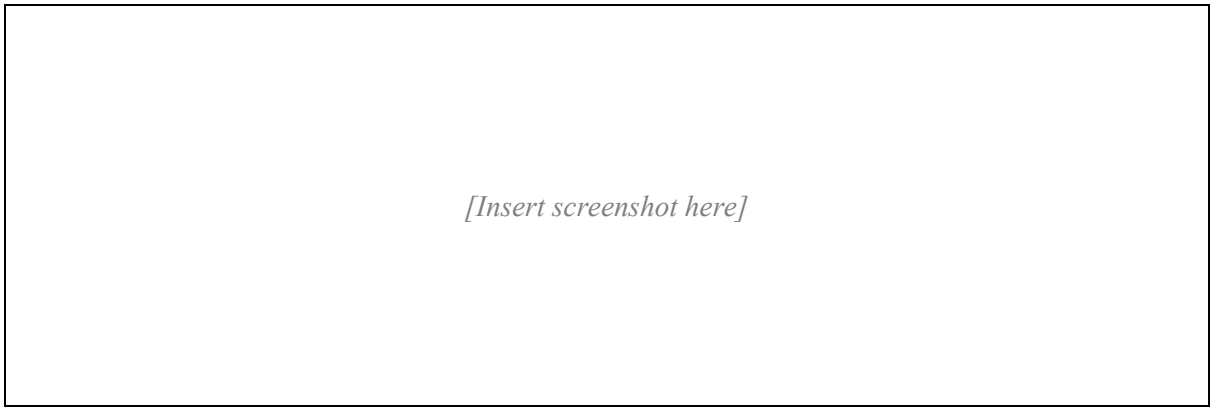


Figure 39: Processing

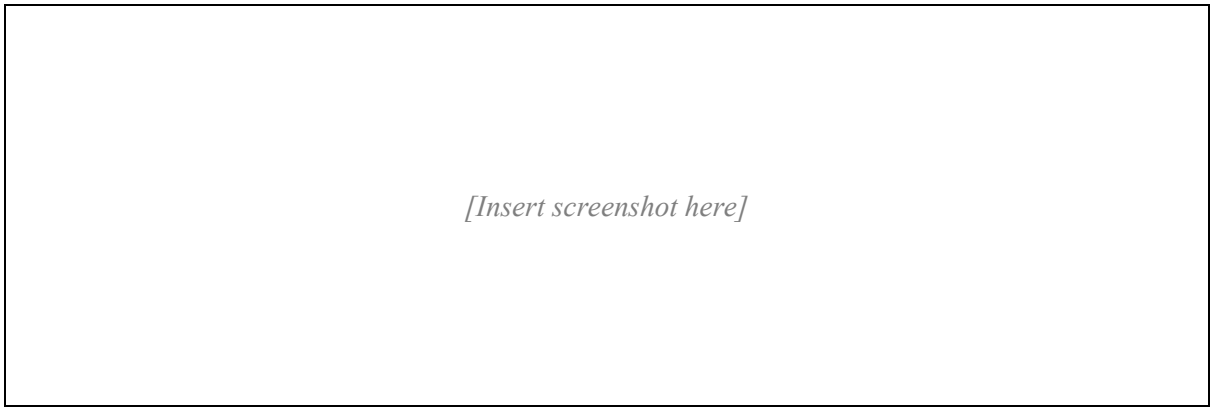


Figure 40: Output

Improvements

[Changes made after the first review. Add before/after screenshots.]

Improvement	Commit ID	Commit Date	Description
[ ]	*[ ]*	*[ ]*	*[ ]*
[ ]	*[ ]*	*[ ]*	*[ ]*

[Insert screenshot here]

Figure 41: Before

[Insert screenshot here]

Figure 42: After